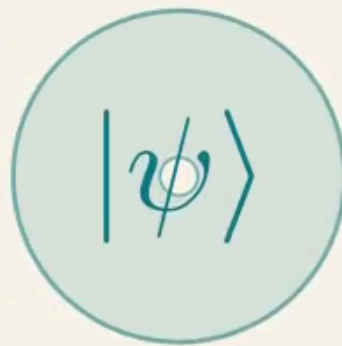




Day 2

Intro to Quantum Computing

KAUST ACADEMY + KGSP

Yesterday: two qubits, one shared thing.

The classical bound was 75%. Entangled qubits hit 85%.

[Bar chart: yesterday's quiz results by concept — CHSH game, Bell histogram, ENIAC parallel]

Today we open the hood.

Yesterday was the feel of it. Today: the actual math —
kept light, kept visual, but real.



$|\psi_1\rangle$



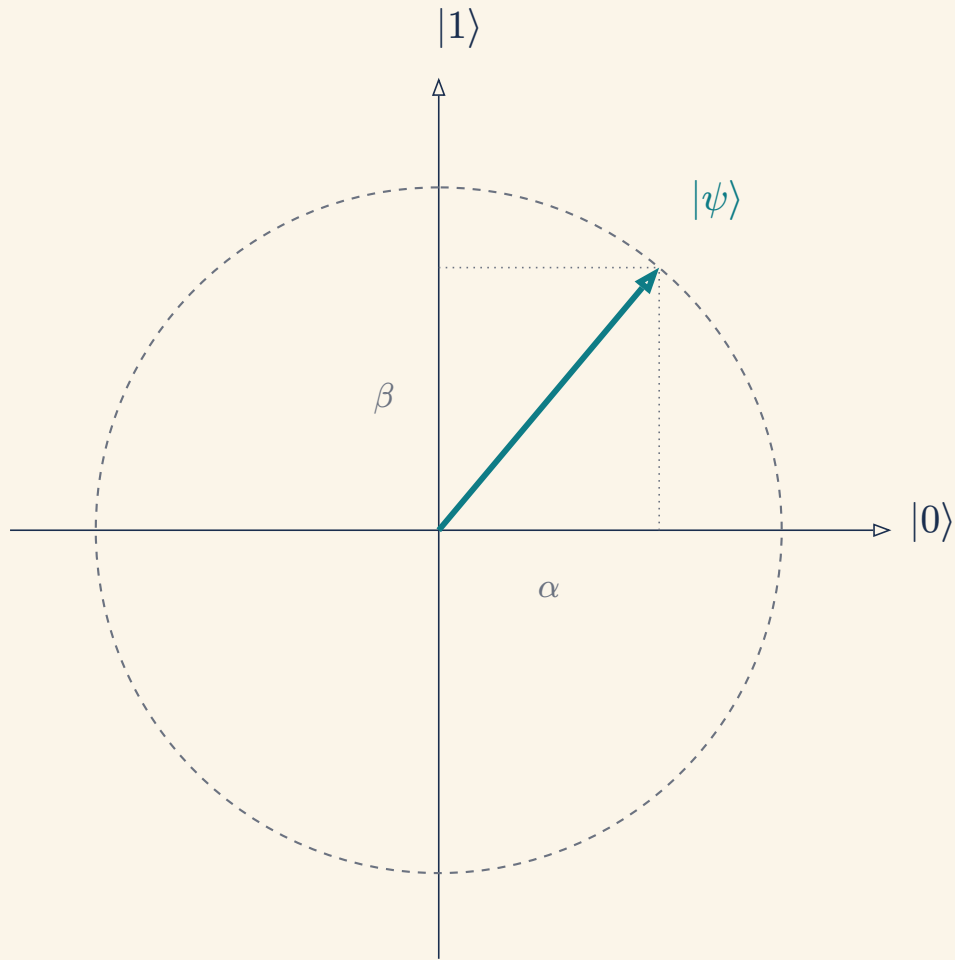
One clear state

Write that pose down. A state is a little **vector** — two slots.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The two definite answers a qubit can give: $|0\rangle$ (“zero”) and $|1\rangle$ (“one”).

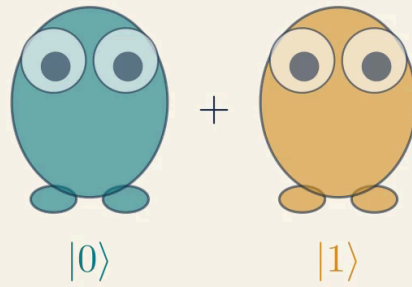
Top slot = amount of $|0\rangle$, bottom slot = amount of $|1\rangle$.



Draw the two slots as two **axes**.

$|0\rangle$ points right, $|1\rangle$ points up. Any state is an arrow to the circle; its shadows on the axes are the amplitudes α , β .

This plane is our home — until the Bloch sphere replaces it.



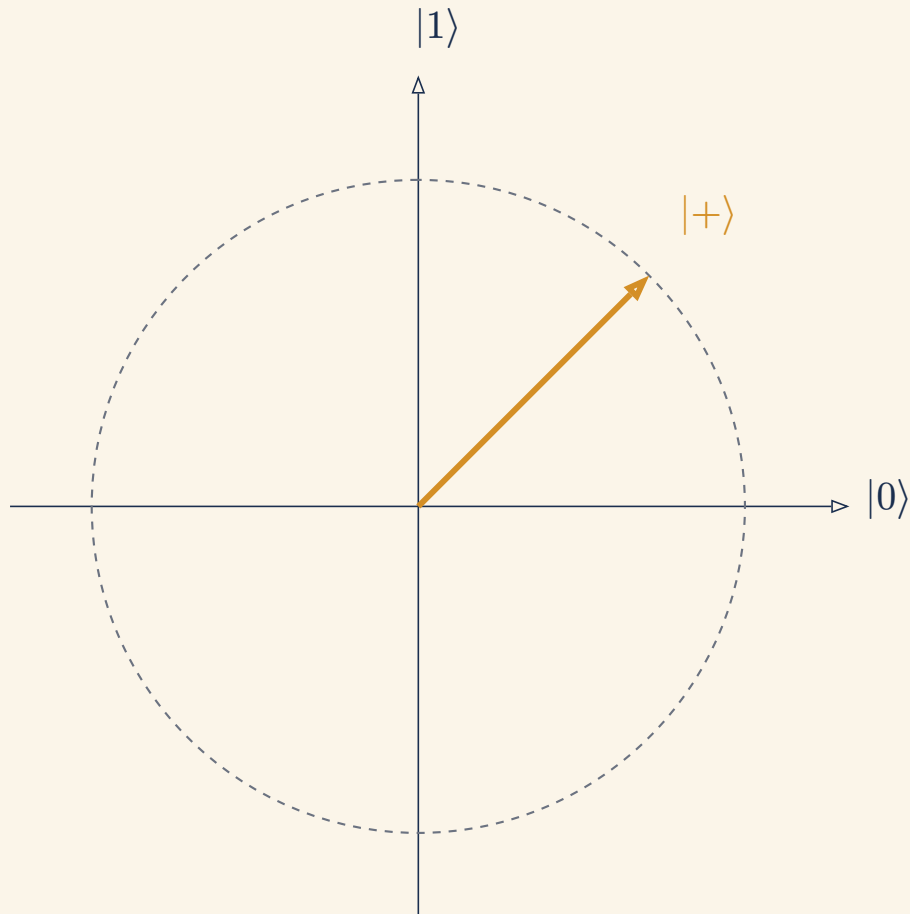
Both, until observed

$[\psi \text{ split-pose from B3}]$

The split pose, written down:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

α, β are the two *amplitudes* — how much of each answer.



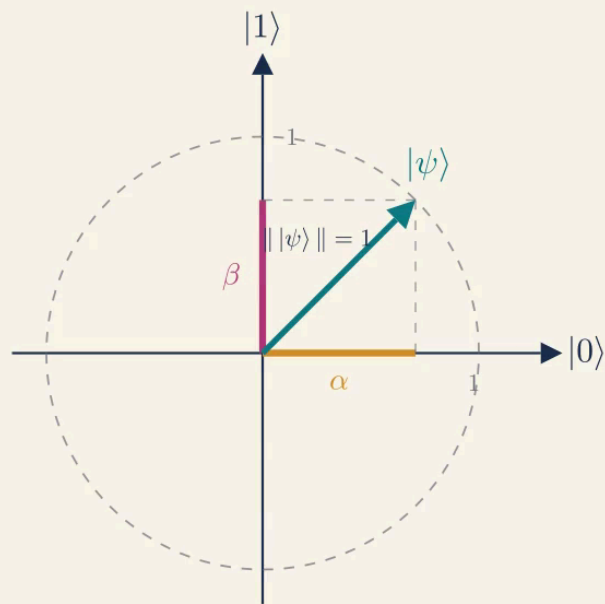
The famous 50/50 mix sits on the diagonal:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

45° — an equal shadow on each axis. (Why $\frac{1}{\sqrt{2}}$, not $\frac{1}{2}$? Next slide.)

A one-slide refresher on probability.

- Each outcome has a probability between 0 and 1.
- Something must happen — the probabilities **sum to 1**.
- Fair coin: $P(H) = P(T) = \frac{1}{2}$, and $\frac{1}{2} + \frac{1}{2} = 1$. ✓



$$\alpha^2 + \beta^2 = 1$$

= “probabilities sum to 1”

the tip rides the circle,

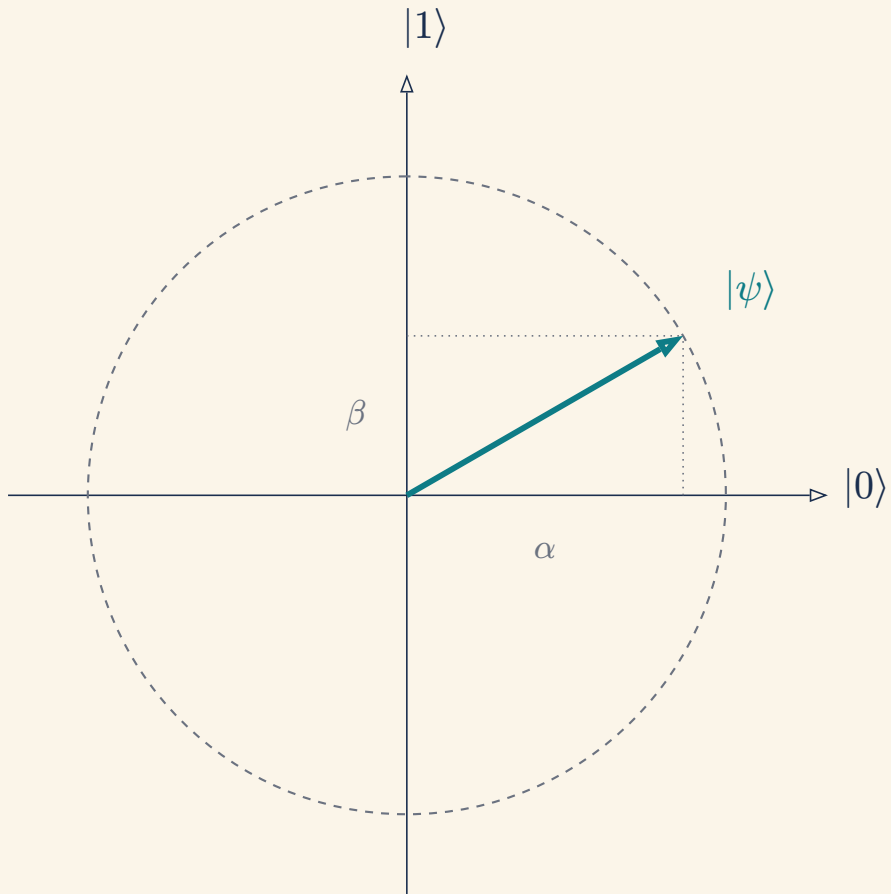
$$\Rightarrow |\alpha| \leq 1, |\beta| \leq 1$$

$$|+\rangle : \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = 1$$

How likely is each answer? Square the amplitude.

$$P(0) = \alpha^2 \quad P(1) = \beta^2$$

$|+\rangle$ gives $P(0) = \frac{1}{2}$, $P(1) = \frac{1}{2}$ — a perfect coin. This is the **Born rule**.



A lopsided state — the arrow leans toward $|0\rangle$:

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$$

$$P(0) = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4} = 75\%, \quad P(1) = \frac{1}{4} = 25\%$$

Longer shadow on $|0\rangle \Rightarrow$ more likely “0”. Yesterday’s 75%, all along.

*A qubit isn't a slower version of a bit.
It's a vector $\alpha|0\rangle + \beta|1\rangle$ — both answers at once,
with α^2 and β^2 the odds you'll see each.*

To ask a question, flip a ket into a **bra** — a row.

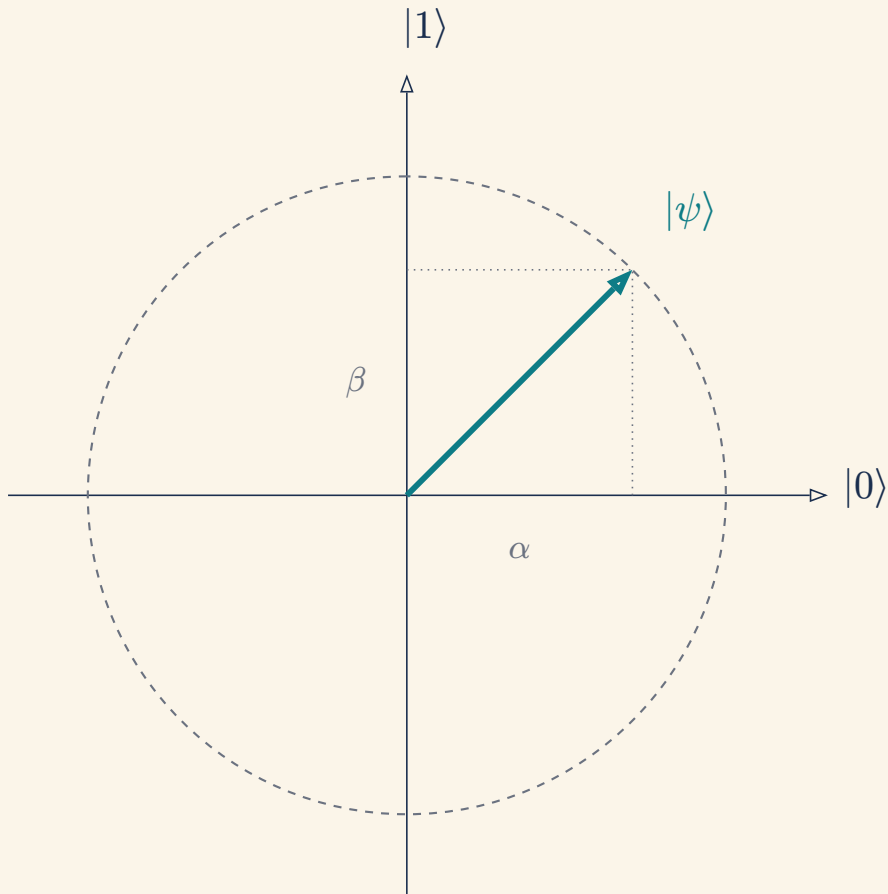
$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \longrightarrow \langle 0| = (1 \ 0)$$

A ket is a state (column); a bra is a question (row). Put them together next.

Bra times ket = one number: the **overlap**.

$$\langle 0|0\rangle = 1 \quad \langle 1|1\rangle = 1 \quad \langle 0|1\rangle = 0$$

$|0\rangle$ and $|1\rangle$ don't overlap at all (0) — a clean pair of axes.



Measuring reads the **shadow**, then squares it.

$$P(0) = \langle 0|\psi\rangle^2 \quad P(1) = \langle 1|\psi\rangle^2$$

The shadow on the $|0\rangle$ axis *is* $\langle 0|\psi\rangle = \alpha$.
 For $|+\rangle$: shadow $\frac{1}{\sqrt{2}}$, squared $= \frac{1}{2}$. Same Born rule
 — now *computed*.



$|0\rangle$

One answer — the rest is gone

The pose, as math:

Before: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

Look $\rightarrow$ land on $|0\rangle$ with prob α^2 , on $|1\rangle$ with prob β^2 .

After: the other slot is *gone* — the vector is now just $|0\rangle$ (or $|1\rangle$).

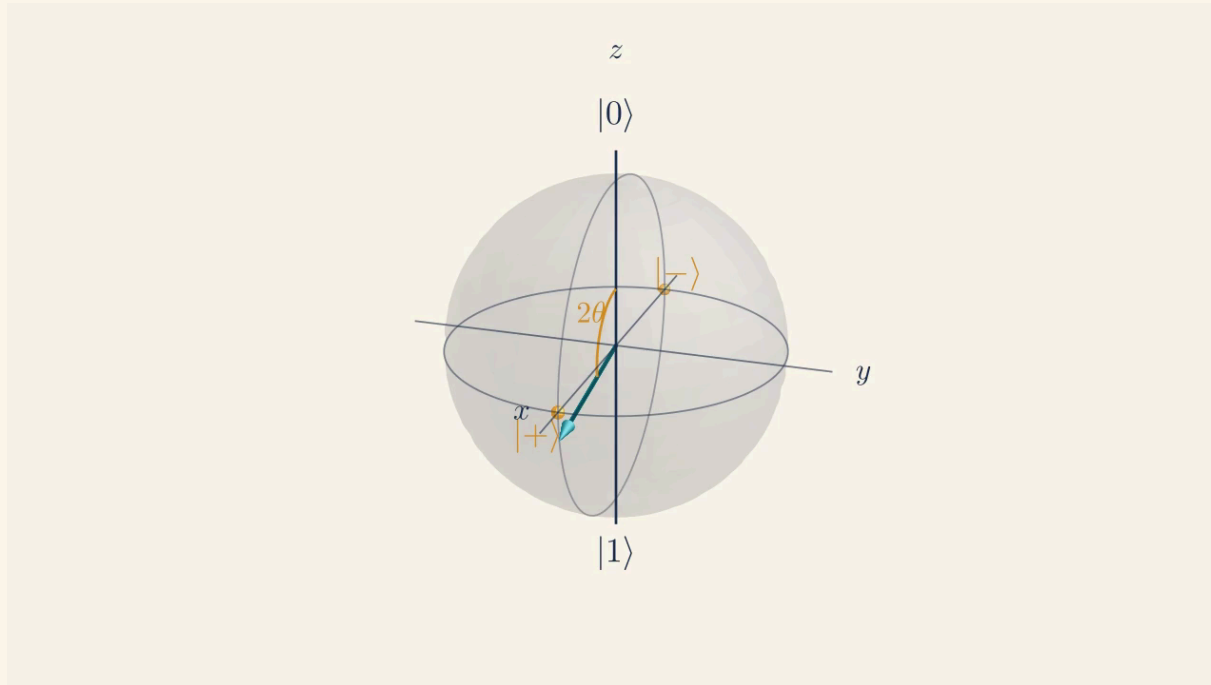
Once you look, there's no going back.

The other amplitude is gone. You can't recover it,
and you can't reconstruct it from the answer you got.

*Measurement isn't reading a hidden value.
It projects ψ onto one answer — and the odds are the overlap, squared.*

Where does ψ *live*?

Every length-1 two-slot vector is a point on one picture.



On the front slice, the tilt just splits the amplitudes:

$|0\rangle$ at the top $|1\rangle$ at the bottom $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ on the equator

More tilt toward the bottom = bigger β = more chance of “1”.

Spinning ψ *around* the axis is the one thing real numbers can't capture.

That go-around angle is the *phase*, and it needs complex amplitudes.
We keep amplitudes real today — just know the equator is a full circle, not two points.

A gate is a **matrix** that turns one state-vector into another.

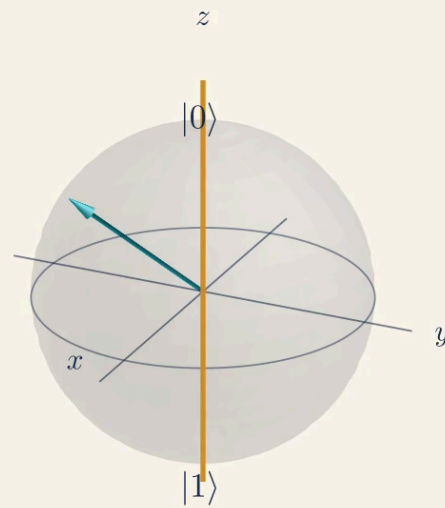
$$|\psi'\rangle = U|\psi\rangle$$

It must keep the length 1 (probabilities still sum to 1) — so every gate is a *rotation* of the sphere.

The three **Pauli gates**: X, Y, Z.

Each is a *half-turn* (180°) about one axis of the Bloch sphere —
X about x , Y about y , Z about z .

$Z - 180^\circ$ about z



Each half-turn, written as a 2×2 matrix:

X — about x

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Y — about y

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Z — about z

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$X|0\rangle = |1\rangle \quad \cdot \text{ bit flip}$$

$$Y|0\rangle = i|1\rangle$$

$$Z|+\rangle = |-\rangle \quad \cdot \text{ phase flip}$$

The i in Y is the complex phase we parked earlier — Y is a flip *and* a phase.

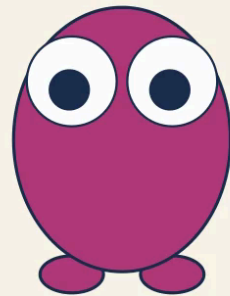
The **Hadamard** gate — not a Pauli.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The gate that *makes* superposition: $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$.

Geometrically, it swings a pole onto the equator.

*Every qubit state is a point on the sphere.
Every gate is a rotation of that point.*



$|\psi_2\rangle$

One qubit had two answers. Two qubits have *four*:

$$|00\rangle \quad |01\rangle \quad |10\rangle \quad |11\rangle$$

So a two-qubit state is a vector with *four* slots, not two.

Combine two single-qubit vectors with the **tensor product** $\otimes$:

$$|0\rangle \otimes |0\rangle = |00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |0\rangle \otimes |1\rangle = |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Multiply each slot of the first by the whole second vector — four numbers out. Slot order: 00, 01, 10, 11.

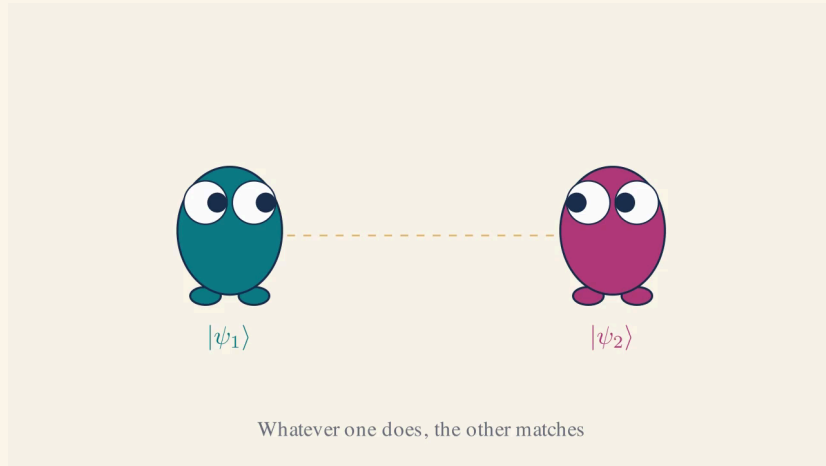
A worked combine: ψ_1 in $|+\rangle$, ψ_2 in $|0\rangle$.

$$|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

This is a *product state*: each qubit still has its own arrow. Hand ψ_1 to Alice, ψ_2 to Bob — cleanly.

The synced-thread pose is one specific four-slot vector:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$



Only 00 and 11 — never 01 or 10. Measure one, you instantly know the other.

Could $|\Phi^+\rangle$ be a product $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$?

That expands to $ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$.

We need $ad = 0$ and $bc = 0$ (no 01, 10) — yet also $ac \neq 0$ and $bd \neq 0$ (keep 00, 11).

Impossible. No such arrows exist — the qubits cannot be separated.

There are exactly four of these.

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \quad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Remember this “four” — it comes back at teleportation.

So how do you *make* a Bell state?

Not with H, X, or Z — each acts on *one* qubit, so it can never correlate two.
You need a gate that **couples** them.

The **CNOT** gate: flip qubit 2 *only if* qubit 1 is $|1\rangle$.

As a 4×4 matrix:

Control $|1\rangle \Rightarrow$ flip the target:

$$|00\rangle \rightarrow |00\rangle$$

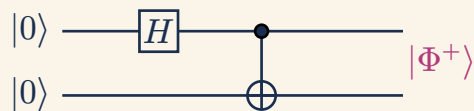
$$|01\rangle \rightarrow |01\rangle$$

$$|10\rangle \rightarrow |11\rangle$$

$$|11\rangle \rightarrow |10\rangle$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

It just swaps the $|10\rangle$ and $|11\rangle$ slots.



Start $|00\rangle$.

After H on qubit 1: $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ — a product state.

After CNOT: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$ — entangled.

No single-qubit gate can entangle two qubits.
It takes a two-qubit gate — H then $CNOT$ turns $|00\rangle$ into a Bell state.
You'll build exactly this circuit after lunch.

One measurement is random. *Many* measurements have a shape.

Run a state N times $\rightarrow$ a histogram. Bar heights approach the amplitude² probabilities.

The *average* outcome (expected value) is what you read off those bars — and what the lab reports.

Yesterday's numbers were probabilities all along.

Classical CHSH: best win probability = $\frac{3}{4} = 75\%$.

Entangled qubits: $\cos^2\left(\frac{\pi}{8}\right) \approx 85\%$ — from amplitudes, squared.

*Two entangled qubits are not two objects.
They are one four-slot vector that no pair of single-qubit arrows can
build.*

You cannot photocopy a qubit.

Not “we haven’t figured out how yet.”

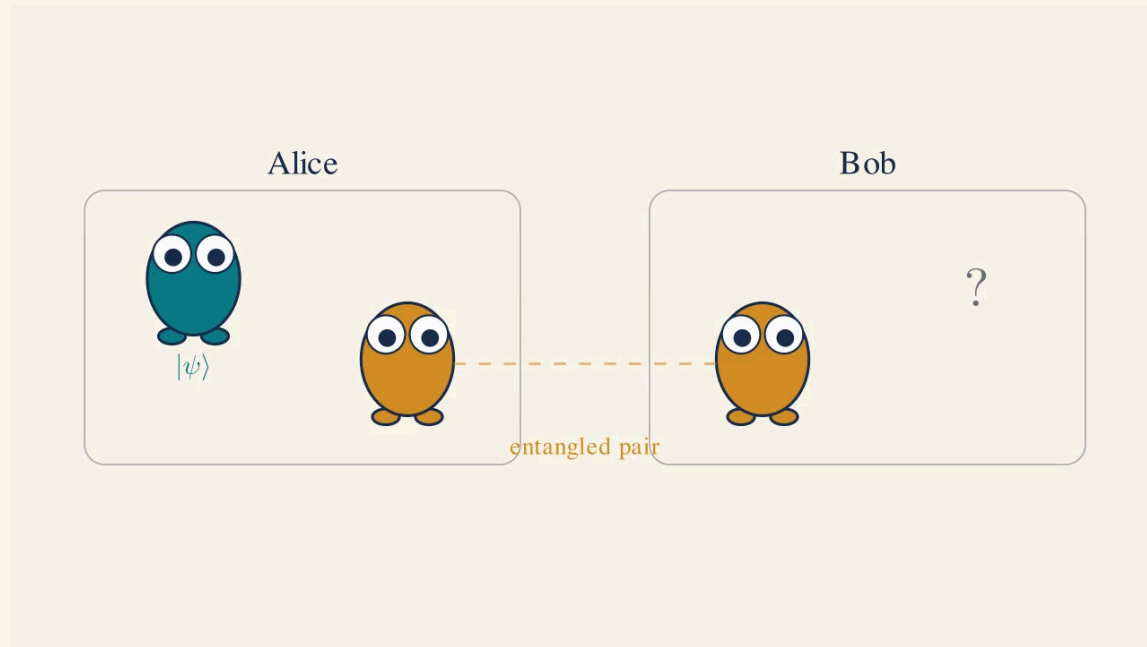
Cannot. Mathematically forbidden.

But Alice has a qubit ψ . And she needs to send it to Bob.

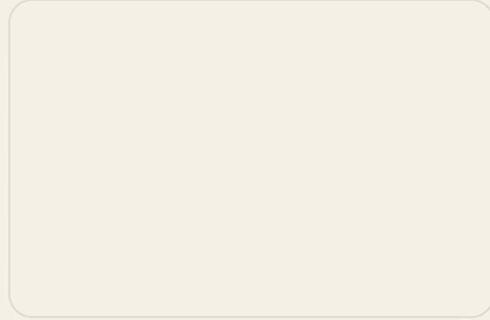
Now what?

Teleportation

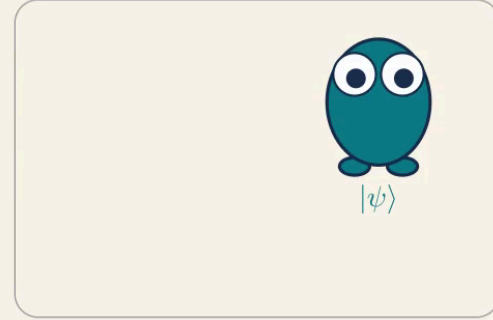
The workaround the universe allows



Alice



Bob



ψ is now Bob's — Alice's is gone

Every count is 4.

The number four is not a coincidence.

Every count is 4.

4

Bell measurement outcomes

The number four is not a coincidence.

Every count is 4.

4

2

Bell measurement outcomes classical bits
(= $\log_2 4$)

The number four is not a coincidence.

Every count is 4.

4

2

4

Bell measurement outcomes

classical bits
($= \log_2 4$)

possible Bob corrections
(Pauli group)

The number four is not a coincidence.

Alice's measurement has 4 possible results.

Two bits are exactly enough to name which of the four.

There are exactly 4 things Bob might do to fix his qubit — one per result.

Every count is 4.

Nothing is arbitrary — the arithmetic writes itself once no-cloning forces Alice to destroy her qubit.

What teleportation ISN'T

What teleportation ISN'T

Not “faster than light.”

Alice's 2 bits travel classically.
Bob is stuck until they arrive.

What teleportation ISN'T

Not “faster than light.”

Alice's 2 bits travel classically.
Bob is stuck until they arrive.

Not “cloning by another name.”

Alice's original is destroyed.
There is never a moment where
two copies exist.

What teleportation ISN'T

Not “faster than light.”

Alice's 2 bits travel classically.
Bob is stuck until they arrive.

Not “cloning by another name.”

Alice's original is destroyed.
There is never a moment where
two copies exist.

Not “Star Trek.”

No matter is transmitted. Only
the **state** moves.

What teleportation ISN'T

Not “faster than light.”

Alice's 2 bits travel classically.
Bob is stuck until they arrive.

Not “Star Trek.”

No matter is transmitted. Only
the **state** moves.

Not “cloning by another name.”

Alice's original is destroyed.
There is never a moment where
two copies exist.

Not exotic.

Done in labs since 1997. You'll
simulate one this afternoon.

After lunch you'll build one.

In Marimo, in Python.

Warmup: quantum coin. Middle: GHZ state.

Showstopper: teleportation.

After lunch

- ✓ Terminal open
- ✓ Repo cloned
- ✓ uv sync done

Anyone snagged:
grab a TA before lunch ends.